Finite-depth Quasi-Phase Matching (QPM) and Converter Efficiency

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An ensemble of ^{85}Rb atoms couples optical (780 nm) and microwave cavities via an **assumed uniform** blue (480/481 nm) control field and UV (297 nm) dressing [1]. Optical-to-microwave conversion absorbs optical and blue photons and emits microwave and UV photons; reversing the loop reverses conversion. A separate 1188 nm lattice creates periodic density for quasi-phase matching (QPM). Finite site width reduces the overlap between the collective atomic profiles. We derive its effect on coherent photon-number conversion efficiency, accounting for atomic loss, reflection, and port coupling.

1. SPATIAL OVERLAP AND LATTICE LOCALIZATION

1.1. Definition and justification of the overlap Q

A collective atomic mode specifies the excitation amplitude and phase on every atom. Let $|j\rangle$ denote a Rydberg excitation on atom j, with all other atoms in the prepared UV-dressed reference state. In a common atomic phase convention, the optical/blue and UV/microwave processes address the normalized states

$$|\psi_u\rangle = \sum_j u_j|j\rangle, \qquad |\psi_v\rangle = \sum_j v_j|j\rangle, \qquad \sum_j |u_j|^2 = \sum_j |v_j|^2 = 1. \tag{1}$$

These are two excitation patterns of the same atoms. The coefficients include relative coupling magnitudes and phases. Their amplitude overlap is

$$Q = |\langle\psi_u|\psi_v\rangle| = |\mathbf{u}^\dagger\mathbf{v}|, \qquad 0 \le Q \le 1. \tag{2}$$

Thus $Q = 1$ means that both processes address the same normalized state, up to an overall phase; $Q = 0$ means that their excitation patterns are orthogonal. The squared overlap Q^2 is the projection weight of one normalized state onto the other. It is neither a fraction of participating atoms nor the full conversion efficiency, which also depends on the coupled dynamics and losses.

For traveling-wave fields with matched uniform envelopes and N atoms,

$$Q = \left|\frac{1}{N}\sum_j e^{i\Delta\mathbf{k}\cdot\mathbf{r}_j}\right| = |\langle e^{i\Delta\mathbf{k}\cdot\mathbf{r}}\rangle|, \tag{3}$$

where the last average is over the normalized atomic density and $\Delta\mathbf{k} = \mathbf{k}_{780} + \mathbf{k}_{\text{blue}} - \mathbf{k}_{\text{UV}} - \mathbf{k}_\mu$. Wavevectors include propagation directions. Each atom contributes a unit phasor: aligned phasors give $Q = 1$, while phase cancellation reduces Q. Exact lattice QPM aligns the phases at the site centres; finite displacements within the wells leave a phase spread and reduce the coherent overlap.

There are many possible collective modes, but only two directly addressed profiles in this model. After choosing the overall phase so that $\mathbf{u}^\dagger\mathbf{v} = Q \ge 0$, write

$$\mathbf{v} = Q\mathbf{u} + s\mathbf{w}, \qquad s = \sqrt{1-Q^2}, \qquad \mathbf{w} = \frac{\mathbf{v} - Q\mathbf{u}}{s}, \qquad \mathbf{u}^\dagger\mathbf{w} = 0, \quad \mathbf{w}^\dagger\mathbf{w} = 1. \tag{4}$$

At $Q = 1$, choose any orthogonal $\mathbf{w}$; its microwave coupling vanishes. The vector $\mathbf{w}$ collects the entire component of $\mathbf{v}$ orthogonal to $\mathbf{u}$ into one normalized pattern. This is a change of basis with no discarded component. The microwave cavity couples to $\mathbf{u}$ with matrix element Qg_m and to $\mathbf{w}$ with $\sqrt{1-Q^2}g_m$. Every remaining pattern can be chosen orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, so neither cavity drives it directly. Section 2 retains both addressed patterns and their optical-excited and Rydberg components.

Indirect population of the remaining modes is excluded only under the assumed uniform internal response. With frozen positions, identical detunings, uniform independent amplitude damping, and uniform blue-control magnitude, the linear atomic dynamics preserve every spatial pattern after the blue phase is absorbed into the atomic basis.

Initially unexcited profiles orthogonal to both addressed patterns then remain unexcited in the coherent response. For example, a common detuning acts as ΔI, so its matrix element between orthogonal profiles is zero. If instead $\Delta_j = \Delta + \delta\Delta_j$, a profile $\mathbf{z}$ orthogonal to both addressed profiles can couple to $\mathbf{u}$ through

$$\mathbf{z}^\dagger \operatorname{diag}(\Delta_j)\mathbf{u} = \sum_j z_j^*\, \delta\Delta_j\, u_j, \tag{5}$$

which need not vanish. Spatial variations in control magnitude or damping can likewise mix profiles. The two-profile reduction is therefore exact for the stated linear model, while its applicability to a device requires checking these assumptions.

Physically, a blue beam much wider than the participating cloud supports a uniform control magnitude. The original transducer derivation assumes uniform detunings and blue control, and reports a blue-beam waist larger than the optical cavity waist [2]. For the present geometry, differential lattice light shifts, UV dressing shifts, and field gradients must also be evaluated. A useful smallness check is $\delta\Delta_{\mathrm{rms}}\tau_{\mathrm{resp}} \ll 1$, with detunings in angular-frequency units and τ_{resp} the relevant response or coherence time; analogous checks apply to variations in control strength and damping. Their effect on the spectrum must be assessed if these checks fail.

The benchmark in Sec. 3 has $\tau_{\mathrm{resp}} = 0.1095\,\mu$s and $\omega_t\tau_{\mathrm{resp}} = 0.0314$, supporting frozen motion on the response timescale when the associated motional phase changes are small. This does not establish uniform detunings or control strengths. Deeper trapping improves localization but can also increase spatially varying differential light shifts. Consequently, computing Q from the density alone does not validate the reduction: the present single-Q prediction is conditional on these uniformity assumptions. If appreciable mixing occurs, additional collective modes must be retained. The overlap in Eq. (2) remains well defined, but it no longer suffices to characterize the spatial dependence of the converter response.

1.2. Lattice localization and quasi-phase matching

For lattice period $d = \lambda_{\mathrm{L}}/2$, define $k_{\mathrm{L}} = 2\pi/\lambda_{\mathrm{L}}$ and $K = 2k_{\mathrm{L}}$. For lattice depth U, the harmonic approximation near a minimum gives [3]

$$V(x) = U\sin^2(k_{\mathrm{L}}x) \simeq U k_{\mathrm{L}}^2 x^2, \qquad E_{\mathrm{R}} = \frac{\hbar^2 k_{\mathrm{L}}^2}{2m}, \qquad \omega_t = \frac{2\sqrt{UE_{\mathrm{R}}}}{\hbar}. \tag{6}$$

At temperature T, the position variance and density harmonics are

$$\begin{aligned} \sigma_x^2 &= \frac{\hbar}{2m\omega_t}\coth\!\left(\frac{\hbar\omega_t}{2k_{\mathrm{B}}T}\right), \\ b_n &= |\langle e^{inKx}\rangle| = e^{-(nK)^2\sigma_x^2/2} = \exp\!\left[-n^2\sqrt{\frac{E_{\mathrm{R}}}{U}}\coth\!\left(\frac{\sqrt{UE_{\mathrm{R}}}}{k_{\mathrm{B}}T}\right)\right]. \end{aligned} \tag{7}$$

Here n labels the spatial harmonic. Even at $T = 0$, zero-point motion gives $b_n < 1$ for $n \neq 0$; deeper trapping increases the overlap. The harmonic approximation requires localization near a minimum.

Exact matching, $\Delta\mathbf{k} = \sum_i n_i\mathbf{K}_i$, aligns the phases of occupied site centres. Independent displacements within sites then give

$$Q = \prod_i b_{n_i,i}; \qquad Q = b_{2,x}\, b_{2,y} \quad \text{for two second-order axes.} \tag{8}$$

For example, take optical and blue beams along $+x$, UV along $+y$, and $\mathbf{K}_i = K\hat{\mathbf{i}}$. The target $(n_x, n_y) = (2, -2)$ requires

$$k_{780} + k_{\mathrm{blue}} - k_{\mu,x} = 2K, \qquad -k_{\mathrm{UV}} - k_{\mu,y} = -2K, \qquad k_{\mu,z} = 0. \tag{9}$$

Exact matching depends on the wavelengths and beam angles. Neglecting $\mathbf{k}_\mu$, nominal wavelengths of 780, 1188, and 480 (481) nm give $\delta k_x = -0.01017$ (-0.03738) rad/μm. For Gaussian site-centre rms widths L_i, residual mismatch adds the factor $\exp[-\sum_i(\delta k_i L_i)^2/2]$; negligible suppression requires $|\delta k_i|L_i \ll 1$. Site factors and mode envelopes must also be evaluated for the actual geometry. The benchmark below assumes exact matching.

2. SIX-MODE CONVERTER RESPONSE

2.1. Collective modes and assumptions

Let $\hat{a}, \hat{b}$ annihilate optical and microwave cavity photons, and let $\hat{e}_j^\dagger, \hat{r}_j^\dagger$ create optical and Rydberg excitations from the prepared UV-dressed state on atom j. UV dressing mixes the ground and auxiliary Rydberg states; the couplings and profiles include this dressing. Weak excitation makes these operators approximately bosonic. Assume frozen positions, uniform blue-control magnitude, identical atomic detunings, and uniform independent amplitude damping. Absorb the blue phase into $\hat{r}_j$ and the global profile phase into $\hat{b}$, so Ω is real and $\mathbf{u}^\dagger\mathbf{v} = Q \geq 0$.

Using the profiles in Eq. (4), with $s = \sqrt{1-Q^2}$, the four collective operators are

$$\begin{aligned} \hat{P}^\dagger &= \sum_j u_j \hat{e}_j^\dagger, \quad & \hat{R}^\dagger &= \sum_j u_j \hat{r}_j^\dagger, \\ \hat{P}_\perp^\dagger &= \sum_j w_j \hat{e}_j^\dagger, \quad & \hat{R}_\perp^\dagger &= \sum_j w_j \hat{r}_j^\dagger. \end{aligned} \tag{10}$$

Uniform control, detunings, and damping preserve this two-profile subspace, so only these four atomic modes and two cavities are driven. Their coupling chain is

$$a \overset{g_o}{\longleftrightarrow} P \overset{\Omega}{\longleftrightarrow} R \overset{Qg_m}{\longleftrightarrow} b \overset{sg_m}{\longleftrightarrow} R_\perp \overset{\Omega}{\longleftrightarrow} P_\perp. \tag{11}$$

The orthogonal pair is a distinct excitation pattern of the same atoms. It is dark to the optical cavity but couples to the microwave cavity and blue field; atomic decay makes its response dissipative.

The interaction Hamiltonian fixes the coupling convention:

$$\frac{\hat{H}_{\mathrm{int}}}{\hbar} = -\Big[g_o \hat{a}^\dagger \hat{P} + \Omega(\hat{P}^\dagger \hat{R} + \hat{P}_\perp^\dagger \hat{R}_\perp) + g_m \hat{b}^\dagger (Q\hat{R} + s\hat{R}_\perp) + \mathrm{h.c.}\Big]. \tag{12}$$

The collective matrix elements g_o, g_m are held fixed as Q varies. The notebooks use $g_o = G/\sqrt{2}$, $g_m = G_\mu/\sqrt{2}$ [4]; these factors reflect conventions, not finite-depth corrections. Here Ω is the Hamiltonian matrix element; Rabi frequencies defined with an extra $1/2$ require conversion. All rates, detunings, and frequencies are angular frequencies.

2.2. Damping and driven equations

Let κ_o, κ_m be bare cavity energy decay rates, including ports and internal loss; useful port rates obey $\kappa_{j,\mathrm{ex}} \leq \kappa_j$. Atomic amplitude decay rates are $\Gamma/2, \gamma_r/2$. Additional microwave energy decay $\gamma_{m,\mathrm{f}}$ includes loss from the lower Rydberg state.

Using Fourier amplitudes $e^{+i\omega t}$ and zero optical-cavity detuning, define

$$\begin{aligned} D_a &= \kappa_o/2 + i\omega, & D_e &= \Gamma/2 + i(\omega - \Delta_e), \\ D_r &= \gamma_r/2 + i(\omega - \Delta_r), & D_b &= (\kappa_m + \gamma_{m,\mathrm{f}})/2 + i(\omega - \Delta_\mu). \end{aligned} \tag{13}$$

Lowercase variables denote the corresponding mean amplitudes. For optical input a_{in} with photon flux $|a_{\mathrm{in}}|^2$, the response is

$$\begin{aligned} D_a a - i g_o p &= i\sqrt{\kappa_{o,\mathrm{ex}}}\, a_{\mathrm{in}}, \\ D_e p - i g_o a - i\Omega r &= 0, \\ D_r r - i\Omega p - i g_m Q b &= 0, \\ D_b b - i g_m (Q r + s r_\perp) &= 0, \\ D_r r_\perp - i\Omega p_\perp - i g_m s b &= 0, \\ D_e p_\perp - i\Omega r_\perp &= 0. \end{aligned} \tag{14}$$

With no microwave input, $b_{\mathrm{out}} = \sqrt{\kappa_{m,\mathrm{ex}}}\, b$ up to a port phase. The transfer amplitude $S_{mo} = b_{\mathrm{out}}/a_{\mathrm{in}}$ gives the coherent photon-number efficiency $\eta = |S_{mo}|^2$, not energy efficiency or added noise.

2.3. Atomic feedback and transfer function

Set $F = D_e D_r + \Omega^2$. Eliminating the orthogonal pair gives

$$r_\perp = \frac{i g_m s D_e}{F}\, b, \qquad \Sigma_\perp = \frac{g_m^2 (1-Q^2) D_e}{F}. \tag{15}$$

This changes D_b to $D_b + \Sigma_\perp$, with frequency-dependent energy-decay contribution $2\,\mathrm{Re}\,\Sigma_\perp$ and dispersion $\mathrm{Im}\,\Sigma_\perp$. This loading does not contribute to the useful output-port rate.

Eliminating all atomic amplitudes gives the exact frequency-domain response

$$\begin{pmatrix} A & iJ \\ iJ & B \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} i\sqrt{\kappa_{o,\mathrm{ex}}}\, a_{\mathrm{in}} \\ 0 \end{pmatrix}, \qquad A = D_a + \frac{g_o^2 D_r}{F}, \quad B = D_b + \frac{g_m^2 D_e}{F}, \quad J = \frac{Q g_o g_m \Omega}{F}, \tag{16}$$

without assuming fast atomic relaxation. The transfer function is

$$\boxed{S_{mo}(\omega) = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, J}{AB + J^2}, \qquad \eta(\omega) = |S_{mo}(\omega)|^2.} \tag{17}$$

At fixed couplings and linewidths, A, B are independent of Q, whereas $J \propto Q$: the useful and orthogonal contributions to microwave loading sum as $Q^2 + (1-Q^2) = 1$. Replacing g_m by $Q g_m$ incorrectly reduces the total loading as well as the cross coupling. At $Q = 1$ the orthogonal pair decouples; at $Q = 0$ conversion vanishes.

2.4. On-resonance efficiency

For comparison with the notebooks, define

$$C_o = \frac{2G^2}{\kappa_o \Gamma}, \qquad C_m = \frac{2G_\mu^2}{(\kappa_m + \gamma_{m,\mathrm{f}})\gamma_r}, \qquad C_B = \frac{4\Omega^2}{\Gamma \gamma_r}, \qquad \eta_j = \frac{\kappa_{j,\mathrm{ex}}}{\kappa_j}. \tag{18}$$

At $\omega = \Delta_e = \Delta_r = \Delta_\mu = 0$, Eq. (17) gives

$$\eta(0) = \eta_o \eta_m \frac{\kappa_m}{\kappa_m + \gamma_{m,\mathrm{f}}} \frac{4Q^2 C_o C_m C_B (1+C_B)^2}{[(1+C_B+C_o)(1+C_B+C_m) + Q^2 C_o C_m C_B]^2}. \tag{19}$$

The non-QPM model in `Abhishek_Converter.nb` assumes fully overlapping collective profiles. Finite overlap modifies both the numerator and denominator of Eq. (19); setting $Q = 1$ recovers the non-QPM efficiency for the same couplings, losses, and port fractions.

Port fractions multiply efficiency at fixed total linewidths; changing port coupling requires updated cooperativities. Propagation and detection losses further reduce this port-to-port efficiency.

3. NUMERICAL BENCHMARK AND LOSS BUDGET

3.1. Lattice and motional scales

Take ^{85}Rb, $\lambda_{\mathrm{L}} = 1188$ nm, waist $w = 200\,\mu$m, and full retroreflection. Adopt the calibration 3 W per beam per axis $\leftrightarrow U/k_{\mathrm{B}} = 15\,\mu$K, without a polarizability calculation. For $E_{\mathrm{R}}/k_{\mathrm{B}} = 0.0799004\,\mu$K, $T = 0$, and two exactly matched second-order axes,

$$b_2 = e^{-4\sqrt{E_{\mathrm{R}}/U}} = 0.746816, \qquad Q = b_2^2 = 0.557734, \qquad Q^2 = 0.311067. \tag{20}$$

The lattice has $U/E_{\mathrm{R}} = 187.7$, $\sigma_x = 36.12$ nm, and $2K\sigma_x = 0.7641$. Thus a deep lattice still leaves substantial phase spread at the second harmonic; the overlap reductions multiply across the two axes. The trap frequency is $\omega_t/(2\pi) = 45.62$ kHz, with $\hbar\omega_t/k_{\mathrm{B}} = 2.190\,\mu$K. The ground-state benchmark requires a temperature well below this scale; at $T = 1\,\mu$K the same harmonic formula gives $Q = 0.48137$.

Treating this density as static requires unresolved motional sidebands and negligible Doppler and recoil phases during τ_{resp}. The Gaussian average retains the full phase spread $nK\sigma_x$; the calculation describes the coherent response, not the inelastic motional spectrum.

3.2. Conversion efficiency

Use zero detunings, $\gamma_{m,\mathrm{f}} = \gamma_r$, and

$$\frac{(G, G_\mu, \Omega)}{2\pi} = (15.8474565,\ 2.09170145,\ 5.52497339)\,\mathrm{MHz}, \qquad \frac{(\kappa_o, \kappa_m, \Gamma, \gamma_r)}{2\pi} = (10,\ 5,\ 6,\ 0.1)\,\mathrm{MHz}. \tag{21}$$

Dividing every rate and frequency by 2π leaves efficiency unchanged. Figure 1 compares the unit-port spectra. The peak efficiency falls from 82.0957% at $Q = 1$ to 47.0824% at $T = 0$ and 38.15% at $T = 1\,\mu$K. Naive Q^2 scaling instead gives 25.5373% at $T = 0$.

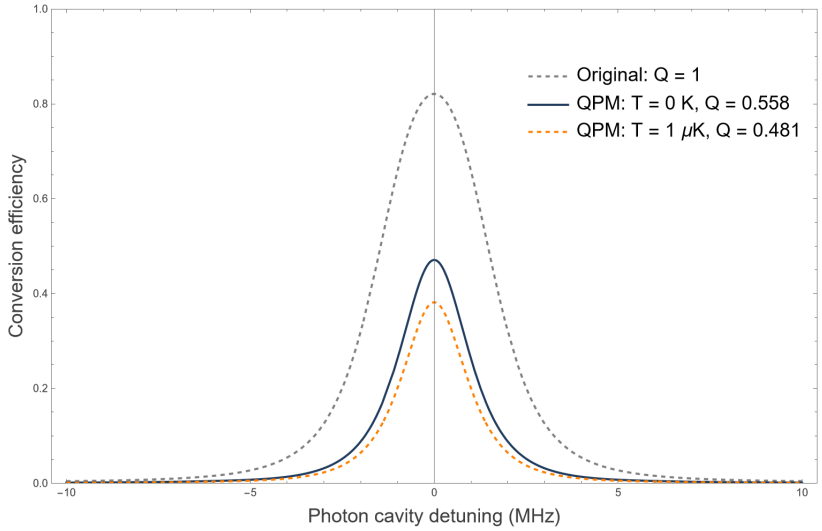


FIG. 1. Conversion spectra from Eq. (17) versus $\omega/(2\pi)$, with parameters in Eq. (21), unit port fractions, and $U/k_\mathrm{B} = 15\,\mu$K per axis. QPM overlaps follow Eqs. (7)–(8).

At $T = 0$ and resonance, $\Sigma_\perp/(2\pi) = 0.147394$ MHz gives an energy-decay contribution $2\,\mathrm{Re}\,\Sigma_\perp/(2\pi) = 0.294787$ MHz. The dressed responses are $A/(2\pi) = 5.20468$ MHz and $B/(2\pi) = 2.76394$ MHz. At $T = 0$, changing only the port fractions to $\eta_o = \eta_m = 0.98$ gives 45.2180%; setting only $\gamma_{m,\mathrm{f}} = 0$ gives 48.54% at unit ports.

The slowest amplitude decay rate of the six-mode drift matrix is $2\pi \times 1.4533$ MHz, so $\tau_\mathrm{resp} = 0.1095\,\mu$s and $\omega_t \tau_\mathrm{resp} = 0.0314$. Combined with the site width, this supports frozen motion; a narrower response requires rechecking these conditions.

3.3. Reflection and dissipation

With the drive convention of Eq. (14), $a_\mathrm{out} = a_\mathrm{in} + i\sqrt{\kappa_{o,\mathrm{ex}}}\,a$. For incident flux $\Phi = |a_\mathrm{in}|^2$, define

$$\begin{aligned} \mathcal{R}_o &= \left|1 + i\sqrt{\kappa_{o,\mathrm{ex}}}\,a/a_\mathrm{in}\right|^2, \quad \mathcal{L}_\mathrm{at} = \frac{\Gamma(|p|^2 + |p_\perp|^2) + \gamma_r(|r|^2 + |r_\perp|^2)}{\Phi}, \\ \mathcal{L}_o &= \frac{(\kappa_o - \kappa_{o,\mathrm{ex}})|a|^2}{\Phi}, \qquad \mathcal{L}_m = \frac{(\kappa_m - \kappa_{m,\mathrm{ex}} + \gamma_{m,\mathrm{f}})|b|^2}{\Phi}. \end{aligned} \tag{22}$$

The passive model obeys $\mathcal{R}_o + \eta + \mathcal{L}_\mathrm{at} + \mathcal{L}_o + \mathcal{L}_m = 1$. On resonance, the unit-port flux budget (percent) is

Channel	$Q = 1$	$Q = 0.557734$
Microwave output	82.0957	47.0824
Optical reflection	3.4748	37.3236
Total atomic dissipation	12.7875	14.6524
Added microwave dissipation	1.6419	0.9416

With mismatched profiles, the orthogonal pair dissipates 2.7759%, included in the atomic total. Of the 35.0133-percentage-point drop in conversion, 33.8487 points appear as increased reflection. Mode mismatch mainly increases reflection here; the conversion penalty is neither $1 - Q^2$ nor the orthogonal pair's dissipation.

4. VALIDITY AND NOTEBOOK CONVENTIONS

Anharmonic motion, resolved sidebands, inhomogeneous controls or detunings, and additional independent optical channels require an extended model. The notebooks and direct matrix solutions verify the six-mode reduction, $Q = 1$ limit, benchmark, and flux conservation; they do not validate the geometry.

In `Abhishek_Converter_QPM.nb`, set $\epsilon = 1$ and $\gamma_{m,\mathrm{f}} = \gamma_r$ (the latter is implicit). Its κ, κ_μ denote κ_o, κ_m; the calculator's $G_{\mathrm{opt}}, G_{\mathrm{mm}}$ denote G, G_μ. The notebook uses x, z for the two axes called x, y here. Since `ConverterEfficQ` retains its forcing amplitude `Ein`,

$$S_{mo} = \sqrt{\eta_o \eta_m}\,\frac{\texttt{ConverterEfficQ}}{\texttt{Ein}}, \qquad \eta = \eta_o \eta_m\,\texttt{NormalizedEfficiencyQ}. \tag{23}$$

For real nonzero drive, `NormalizedEfficiencyQ` = `EfficiencyQ`/`Ein`^{2}. `EfficiencyQWithPorts` multiplies the unnormalized expression, so use `Ein=1` or divide its result by `Ein`^{2}. `SigmaPerp` and `OnResonanceEfficiencyQ` implement Eqs. (15) and (19), the latter at unit ports. Use `parmsScreenshot` for Eq. (21); `parmsOriginal` retains the original parameters. Keep Q symbolic until applying `qRule`.

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[1] A. Kumar *et al.*, Nature **615**, 614 (2023).

[2] A. Kumar *et al.*, Quantum-limited millimeter wave to optical transduction (2022), supplement B.1–B.3, arXiv:2207.10121.

[3] I. Bloch, J. Dalibard, and W. Zwerger, Rev. Mod. Phys. **80**, 885 (2008), sec. II.

[4] (2026), baseline: `Abhishek_Converter.nb`; QPM implementation: `Abhishek_Converter_QPM.nb`. Code and calculator: AVKGitHub1/transduction_calcs, commit `ca24030`. The Q extension is derived here.